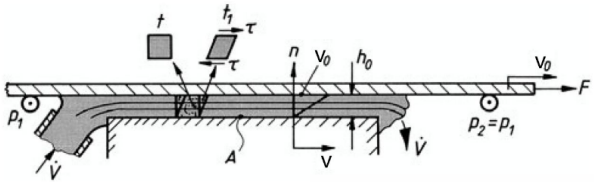
# I. Viscous flow

## Symbology

| Symb | Name | Units | Symb | Name | Units |
|---|---|---|---|---|---|
| $\nu$ | Kinematic viscosity | m$^2$/s | $\eta$ | Dynamic viscosity | Pa·s |
| $\tau$ | Shear stress | Pa | $dn$ | Normal distance | m |
| $du/dy$ | Rate of strain | [-] | $e_{\text{Diss}}$ | Spec. diss. power | W/m$^3$ |

## 1. Flows with friction




$$F = \eta \frac{v_0 \cdot A}{h_0} \implies \tau = \frac{F}{A} = \eta \frac{v_0}{h_0} = \eta \frac{dv}{dn}$$

For pipes:

$$F_\tau = \tau A \implies M_\tau = F_\tau y = \tau \cdot l \cdot 2\pi r_0^2$$

### 1.1 Friction law

$$\tau = -\eta \frac{dv}{dn} \quad ; \quad \dot{e}_{\text{Diss}} = \eta \left(\frac{dv}{dn}\right)^2$$

## 2. Viscosity

If $\eta = 0$, the fluid is called inviscid.

No slip condition: velocity of a particle closest to the wall has velocity equal to the wall velocity.

### 2.1 Reynolds Number

$$Re = \frac{\rho v L}{\eta} = \frac{vL}{\nu} \quad ; \quad \nu = \frac{\eta}{\rho}$$

### 2.2 Newtonian fluids

- Rate of deformation $\mathrm{d}u/\mathrm{d}y$ (velocity gradient) is linearly proportional to the shear stress $\tau$
- The constant of proportionality is the viscosity (slope)
- $\eta = \dfrac{\tau}{\mathrm{d}u/\mathrm{d}y} = \text{constant}$

### 2.3 Non-Newtonian fluids

Non-linear relation between shear stress and deformation rate due to non-constant viscosity.

$$\tau = \eta \left(\frac{\partial u}{\partial y}\right) \cdot \frac{\partial u}{\partial y}$$

#### 2.3.1 Dilatant (shear thickening) fluids

Viscosity increases with increasing strain rate.

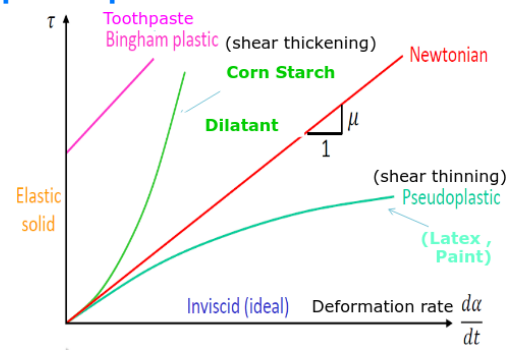
#### 2.3.2 Plastic or pseudo-plastic (shear thinning) fluids

Viscosity decreases with increasing strain rate.

#### 2.3.3 Bingham medium

Flow occurs only after a yield stress $\tau_0$ is reached.

### 2.4 Graphical representation



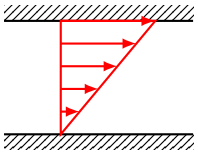

## 3. Flow profiles

### 3.1 Couette flow profiles (constant velocity gradient)

$$\frac{\partial u(y)}{\partial y} = \frac{v_p}{s}$$

$$e_{\text{Diss}} = \eta \left(\frac{\partial u(y)}{\partial y}\right)^2 = \eta \left(\frac{v_p}{s}\right)^2$$

$$P_{\text{Diss}} = \int_V e_{\text{Diss}}(y)\, dV$$



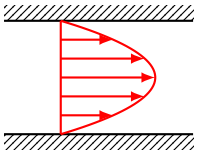
### 3.2 Poiseuille flow profiles (parabolic velocity gradient)

$$\frac{\partial u(y)}{\partial y} = \frac{4v_{\text{max}}}{s}\left(1 - \frac{2y}{s}\right)$$

$$v(r) = ar^2 + br + c$$

$$e_{\text{Diss}} = \eta \frac{16 v_{\text{max}}^2}{s^2}\left(1 - \frac{2y}{s}\right)^2$$

$$P_{\text{Diss}} = \frac{16\eta A v_{\text{max}}^2}{3s}$$



## 4. Laminar pipe flow

### 4.1 $p_1 > p_2$

$$\frac{\partial p}{\partial z} = \frac{p_2 - p_1}{L}$$

### 4.2 Constant linear flow profiles

$$\dot{V} = v_m A = v_m R^2 \pi$$

### 4.3 Parabolic flow profiles

$$\dot{V} = \int v(r)\, dA = \frac{v_{max} \cdot R^2 \pi}{2}$$

$$v(r) = v_{max}\left(1 - \frac{r^2}{R^2}\right)$$

$$\frac{dv}{dr} = -2v_{max}\frac{r}{R^2}$$

$$P_{\text{Diss}} = \int_0^L \int_0^{2\pi} \int_0^R \eta \left(-2v_{max}\frac{r^2}{R^2}\right) \cdot r\, dr\, d\varphi\, dx$$

$$P_{\text{Diss}} = 2\pi L \int_0^R \eta \left(-2v_{max}\frac{r^2}{R^2}\right) \cdot r\, dr = 2\pi L \eta v_{max}^2 = 8\pi L \eta v_m^2$$

## 5. Flows in gaps and bearings

### 5.1 Assumptions in gaps and bearings

- Newtonian fluid with constant viscosity $\eta$
- Incompressible fluid
- Very low Reynolds number $Re < 2000$
- Laminar flow
- Balance of pressure and viscous forces
- No slip condition at the walls
- No acceleration (negligible)
- Pressure is uniform across the thickness $\partial p/\partial y \approx 0$

### 5.2 Equilibrium of forces

$$\sum F_x = 0 \implies pA_p - (p + dp)A_p + \tau A_\tau + (\tau + d\tau)A_\tau = 0$$

$$A_p = dy \cdot b \quad ; \quad A_\tau = dx \cdot b \quad ; \quad \tau = \eta \frac{dv}{dy} \to \frac{d\tau}{dy}$$

$$\boxed{\eta \frac{d^2 v(x,y)}{dy^2} = \frac{dp(x)}{dx} = p'}$$

### 5.3 Basic velocity equation

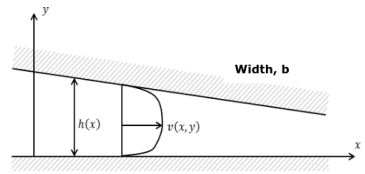
$$v(x,y) = \frac{1}{\eta} \cdot p' \cdot \frac{y^2}{2} + c(x)y + d(x)$$

#### 5.3.1 Gap flow (walls not moving)

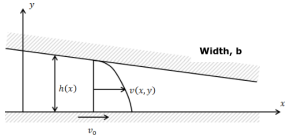
$$v(x,y) = \frac{p'(x)}{2\eta}\left[y^2 - h(x)y\right]$$

$$c(x) = -\frac{h(x)}{2\eta} \cdot \frac{dp(x)}{dx}$$

$$d(x) = 0$$



### 5.3.2 Plain bearing / Creeping gap flow (one wall moving)




Assumption for $Re \ll 1$: $\eta \dfrac{d^2 v}{dy^2} = \dfrac{dp}{dx} = p'$

$$v(x,y) = \frac{p'(x)}{2\eta}\left[y^2 - h(x)y\right] + \frac{v_0}{h(x)}\left[h(x) - y\right]$$

$$c(x) = -\frac{h(x)}{2\eta} \cdot \frac{dp(x)}{dx} - \frac{v_0}{h(x)}$$

$$d(x) = v_0$$

For the simplest, parabolic velocity profile case:

$$h(x) = h_0 \quad ; \quad v_0 = 0 \quad ; \quad p' = \frac{dp(x)}{dx} = \frac{\Delta p}{\Delta L} = -\frac{12\eta v_m}{h_0^2}$$

$$v(y) = \frac{p'}{2\eta}\left(y^2 - h_0 y\right) \Longrightarrow \frac{\partial v}{\partial y} = 0 = \frac{p'}{2\eta}\left(2y - h_0\right)$$

$$y = \frac{h_0}{2} \Longrightarrow v_{max} = v\left(y = \frac{h_0}{2}\right) = -\frac{h_0^2 p'}{8\eta} = \frac{3}{2} v_m$$

$$v_m = -\frac{\Delta p \cdot h_0^2}{12\eta L}$$

$$\dot{V} = b\int_0^{h_0} v(y)\,dy = -b\frac{p' h_0^3}{12\eta} = \frac{h_0^2}{6\eta} b\Delta p$$

General sliding bearings:

$$\dot{V} = b\int_0^{h(x)} \frac{p'}{2}\eta\left(y^2 - h(x)y\right) + \frac{v_0}{h(x)}\left(h(x) - y\right)\,dy$$

$$\dot{V} = -b\frac{p' h_{(x)}^3}{12\eta} + b\frac{v_0 h(x)}{2} = ¢$$

$$p' = 6\eta v_0\left(\frac{1}{h(x)^2} - \frac{\dot{V}}{b h(x)^3}\right)$$

### 5.3.3 General form of the volumetric flow rate

$$v(x,y) = \frac{1}{\eta} p' \frac{y^2}{2} + c(x) + d(x)$$

$$\dot{V} = b\int_0^{h(x)} v(x,y)\,dy = b\left[-\frac{h^3(x)}{6\eta}p' + c(x)\frac{h^2(x)}{2} + d(x)h(x)\right] = ¢$$

$$p' = \frac{6\eta}{h(x)^3}\left(\frac{\dot{V}}{b} - c(x)\frac{h(x)^2}{2} - d(x)h(x)\right)$$

### 5.3.4 Possible semplifications

If h $\ll$ D:

$$A = \int_A r\,dr\,d\varphi \approx \pi D h$$

## 5.4 Specific dissipation power

$$P = \tau A v = \eta \frac{\partial v}{\partial n} A v \quad ; \quad dP = dF dv$$

$$dF_\tau = d\tau dA = \eta \frac{dv}{dy} dx dz$$

$$e_{\text{Diss}} = \frac{dP_\tau}{dV} = \frac{\eta \frac{dv}{dy} dx dz dv}{dx dy dz} = \eta\left(\frac{dv}{dy}\right)^2 = \eta\left(\frac{\partial v}{\partial y}\right)^2$$

$$P_{\text{Diss}} = \int_V e_{\text{Diss}}\,dV = \iiint_V e_{\text{Diss}}\,dx dy dz = M \cdot \omega$$

### 5.4.1 Pressure loss according to Bernoulli

$$p_1 - p_2 = \rho \frac{P_{\text{Diss}}}{\dot{m}} = \frac{P_{\text{Diss}}}{\dot{V}} = \frac{8\pi L \eta v_m}{R^2} \; (= \rho g h)$$

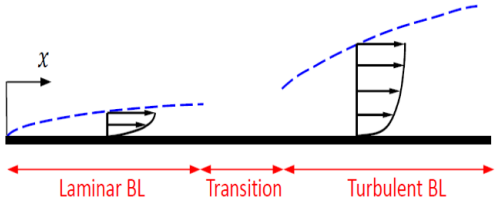
# II. Boundary layers

| Symb | Name | Units | Symb | Name | Units |
|---|---|---|---|---|---|
| $v_\infty$ | free-stream velocity | m/s | $v(y)$ | local velocity in boundary layer | m/s |
| $\delta$ | boundary layer thickness | m | $\delta^*$ | BL displacement thickness | m |
| $\theta$ | momentum thickness | m | $\tau_w$ | wall shear stress | Pa |
| $\nu$ | kinematic viscosity | m²/s | $\eta$ | dynamic viscosity | Pa·s |
| $Re_x$ | Reynolds num. at position $x$ | - | $\dot{V}_{BL}$ | boundary-layer volume flow rate | m²/s |

Boundary layers develop due to wall shear stress $\tau$ and fluid viscosity $\nu$.

## 6. Laminar vs. Turbulent boundary layers

## 6.1 Critical Reynolds number

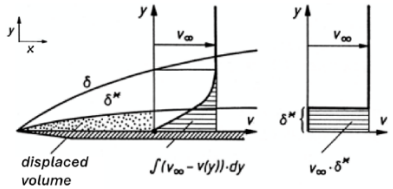



Reynolds number for the BL on a flat plate:

$$Re_x = \frac{v_\infty x}{\nu} = \frac{v_\infty \cdot l_{char}}{\nu}$$

$Re$ **increases** with the distance from the plate entry.
**Transition to turbulent**: $Re_x \geq 5 \times 10^5$

## 6.2 Flat plate boundary layer




$$\delta(x) \approx \sqrt{x} \quad ; \quad \tau(x) \approx 1/\delta(x) = 1/\sqrt{x}$$

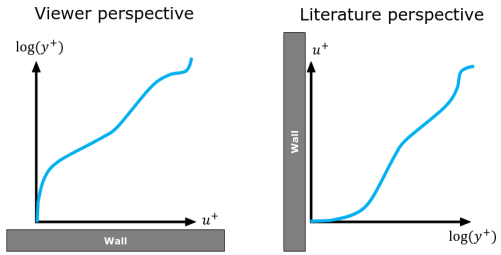
$$\dot{V}_{BL} = v_\infty \cdot \delta^*(x)$$

with 2 plates: $\dot{V}_{BL} = A^* \cdot v = \left(\dfrac{d - 2\delta^*}{2}\right)\pi \cdot v$

## 6.3 Turbolent BL

### 6.3.1 Non-dimensional forms

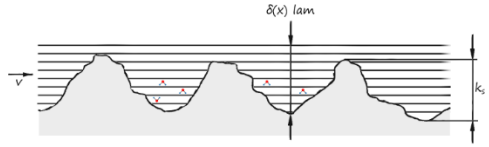
Non-dimensional plot of wall normal turbulent velocity profile:



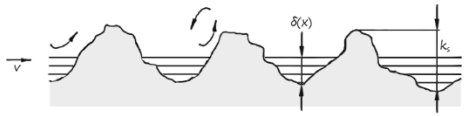

- Friction velocity: $u_\tau = \sqrt{\dfrac{\tau_{\text{wall}}}{\rho}}$
- Dimensionless wall parallel velocity: $u^+ = \dfrac{u}{u_\tau}$
- Dimensionless wall distance: $y^+ = \dfrac{y u_\tau}{\nu}$

## 6.4 Rough and smooth walls and the effect on friction

### 6.4.1 No-slip condition




Hydraulically Smooth Surface




Rough Surface

- Hydraulically smooth surface: $\delta(x) > k_s$
- Rough surface: $\delta(x) < k_s$. Leads to vortex.

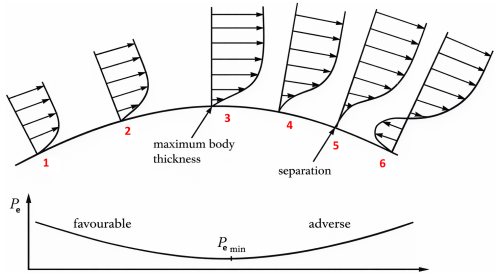
# 7. Flow separation

## 7.1 Flow separation conditions

1. Friction
2. Pressure increase

At sharp edges, the flow always separate.

## 7.2 Pressure gradient



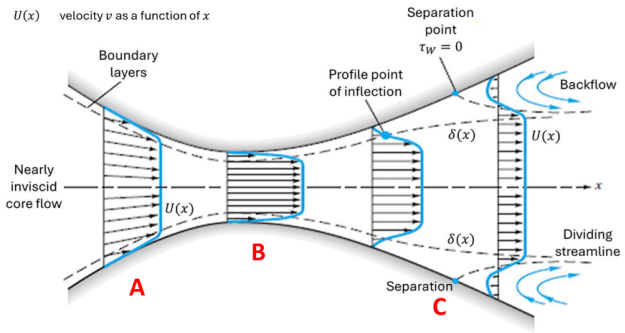

### 7.2.1 Comparison of boundary layer profile shape

Comparing BL as a function of the pressure gradient:

$$\frac{dP}{dx} \sim -\frac{dU(x)}{dx} \quad ; \quad \tau_{\text{wall}} = m\frac{dv}{dy_{y=0}}$$

1. favorable pressure gradient: $\frac{\partial p}{\partial x} \ll 0$
2. favorable pressure gradient: $\frac{\partial p}{\partial x} < 0$
3. zero pressure gradient: $\frac{\partial p}{\partial x} = 0$
4. mild adverse pressure gradient: $\frac{\partial p}{\partial x} > 0$
5. critical adverse pressure gradient: $\frac{\partial p}{\partial x} > 0, \tau_{\text{wall}} = 0$
6. large adverse pressure gradient: $\frac{\partial p}{\partial x} \gg 0$

## 7.3 Flow separation gradients




### 7.3.1 Nozzle (Favorable Pressure Gradient) (A)

$$\frac{dp}{dx} < 0 \quad ; \quad \frac{dU}{dx} > 0$$

- Decreasing pressure and decreasing area
- Pressure decreases in the flow direction
- Pressure force is in the flow direction
- No flow separation

### 7.3.2 Throat (B)

$$\frac{dp}{dx} = 0 \quad ; \quad \frac{dU}{dx} = 0$$

- Constant pressure and constant area
- Fluid particles in the BL slow down due to shear stress only
- No flow separation can occur
- Similar to a flat plate

### 7.3.3 Diffuser (Adverse Pressure Gradient) (C)

$$\frac{dp}{dx} > 0 \quad ; \quad \frac{dU}{dx} < 0$$

- Pressure increases in the flow direction
- Pressure force is in opposite direction of the flow
- **Back flow situation**: Fluid particles close to the wall with low momentum may come to a stop or even move in opposite direction of the main flow

## 7.4 Influence of $Re$ on blunt bodies

- $Re < 1$: No vortex formation
- $Re \approx 10^5$: Laminar BL, vortex formation in dead water area
- $Re > 10^6$: BL partially turbolent, vortex on the body surface

# III. Drag and Lift Forces

**Drag force**: parallel and opposite to the relative flow direction
**Lift force**: perpendicular to the relative flow direction

# 8. Lift force and Drag coefficient

## 8.1 Drag force

$$\boxed{F_D = C_D q_\infty A_P = C_D \frac{1}{2}\rho v_\infty^2 A_p}$$

where:

$C_D$ : drag coefficient $(\mu)$ ; $A_p$ : project area
$q_\infty$ : dynamic contribution

### 8.1.1 Drag on a body in a flow

$$\boxed{F_D = F_P + F_\tau}$$

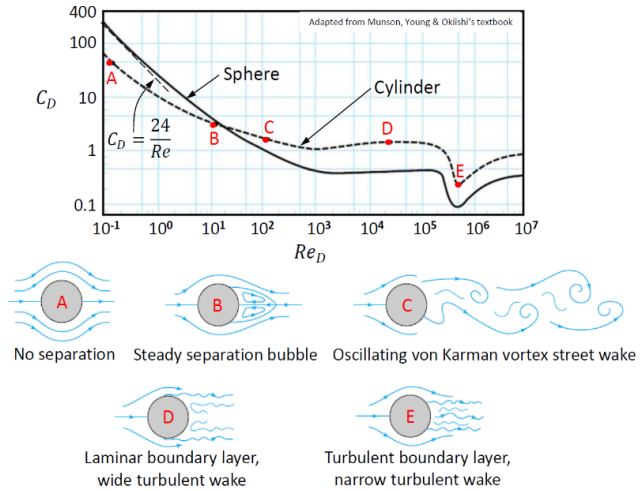
**Friction drag**: friction forces $F_\tau$ due to wall shear stress $\tau_{\text{wall}}$
**Pressure drag**: pressure forces $F_P$ due to pressure variation along the surface of the object

### 8.1.2 Drag power

$$\boxed{P_D = F_D v_\infty}$$

## 8.2 Drag on smooth circular cylinders

### 8.2.1 Prandlt's experiment




### 8.2.2 Stokes formula

$$\boxed{F_D \approx F_\tau = 3\pi\eta d v_\infty}$$